Mingzhe Li

WINTERHACK 2027 APPLICANT

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Information and Computing Science undergraduate with experience in C++ and Python, mathematical modelling, and genetic algorithm optimisation. Interested in applying these skills to multi-agent decision-making and simulation-based engineering challenges.

EDUCATION

BUPT & QMUL STUDENT 2024 - Present

Information and Computing Science

GPA: 3.76/4.00 **Academic rank:** Year 1: 8/97; Year 2: 15/95

WHY I WANT TO PARTICIPATE

I want to apply my mathematics and programming skills to an engineering challenge where ideas must be implemented, tested and improved. Robot soccer interests me because it connects optimisation with decision-making under uncertainty, and requires individual actions to work together as a team strategy.

I also want to learn from students across campuses, compare different approaches and practise communicating technical ideas in English. I would contribute my C++ and Python skills, experience with genetic algorithms and AI coding tools, and willingness to test assumptions and refine solutions with teammates.

PROJECT EXPERIENCE AND AWARDS

Genetic Algorithm for Rectangular Part Nesting

- Carried out the main development work on a genetic algorithm for optimising the layout of rectangular parts that can be joined together.
- Applied optimisation methods to a geometric layout problem, combining mathematical modelling with algorithm implementation.

Honorable Mention | Mathematical Contest in Modeling (MCM), 2025

Silver Medal | 18th BUPT Programming Contest, 2024

SKILLS

Programming: C++, Python

Methods: Mathematical modelling, optimisation algorithms, genetic algorithms

AI tools: AI-assisted coding and agent development

Language: English: strong proficiency